

CP-322-Assignment 1 Q2

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Decision Trees are a machine learning method used to classify data by making a series of decisions based on different features. For this question, we used a decision tree to classify news headlines as either real or fake. We tested different maximum tree depths to see which one would give us the highest accuracy on the 15% validation set. As the depth increased, the accuracy generally improved, starting at 60% when the maximum depth was 2 and reaching 77.35% at a depth of 100. The accuracy stayed the same for depths of 150, 200, and 300, so we chose 100 as the best maximum depth since it was the smallest depth that achieved the highest accuracy. When we tested this model on the separate 15% test set, the accuracy dropped slightly to 75.51%. Lastly, we looked at the first two layers of our best decision tree and found that the first split was based on the word "donald." Depending on whether that word appeared in a headline, the next split checked either "trumps" or "the." This showed us how the decision tree uses individual words to help determine whether a headline is real or fake.

